

Report For Correlation Test Between Two Variables

- 1- Correlation test is used to evaluate the association between two or more variables.
 - 2- From **Shapiro-Wilk normality test** :- the two p-values are Less than the significance level 0.05 implying that the distribution of the data are not from normal distribution. So, we can not assume the normality.
 - 3- From the **normality plots** :- we conclude that both populations not come from normal distributions.
 - 4- **Correlation test** :- The p-value of the test is 0.02348, which is less than the significance level $\alpha = 0.05$. We can conclude that Dite and Age are significantly correlated with a correlation coefficient of 0.110407 and p-value of 0.02348.
 - 5- The correlation coefficient between x and y are 0.110407 and the p-value is 0.02348.
 - 6- If correlation coefficient :-
 - -1 indicates a strong **negative correlation** : this means that every time x **increases**, y **decreases** (left panel figure)
 - 0 means that there is no **association** between the two variables (x and y) (middle panel figure)
 - 1 indicates a strong **positive correlation** : this means that y **increases** with x (right panel figure)
- The Correlation Coefficient in our data between Diet and Age = 0.110407. So, It indicates a weak positive correlation .